

Ankush Das

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Summary

Computer Science graduate with hands-on experience in C++, **backend development**, **full-stack applications**, and **AI-integrated systems**. Experienced in developing APIs, object-oriented applications, and software solutions with a focus on **clean, modular, and maintainable code**.

Technical Skills

Languages: C++ · C · C# · Java · Python · JavaScript · SQL / PL/SQL

Backend: Node.js · Express.js · REST APIs · API Integration

Frontend: React.js · Avalonia UI · AXAML · HTML · CSS

Databases: MongoDB · Oracle

AI / ML: Groq API · OpenAI API · Fast API · NLP · Transformers · Prompt Engineering

Tools: Git · GitHub · Postman · CMake · Gnuplot · Streamlit

Core Concepts: Data Structures & Algorithms · OOP · Operating Systems · Computer Networks · DBMS

Education

Sardar Vallabhbhai National Institute of Technology, Surat

B.Tech in Computer Science and Engineering

Nov 2022 – May 2026

Surat, India

Experience

Member of Technical Staff Intern, CCTech

Pune, India

Jul 2026 – Present

- Developed a C++ **geometry application** for maximum rectangle fitting within an irregular pentagon using OOP and SOLID design principles.
- Implemented reusable geometry components using **STL**, **smart pointers**, **exception handling**, **file I/O**, **CMake**, and **Gnuplot** for computation and visualization.
- Developed a desktop **Paint application using C# and Avalonia UI** with drawing, erasing, color selection, brush-size control, and canvas operations.
- Implemented **pointer-event based drawing**, **Polyline rendering**, and **stack-based Undo/Redo** using AXAML, event handling, and data binding.

Projects

Paint Application

C# · Avalonia UI · AXAML

- Developed a desktop **Paint application using C# and Avalonia UI** with drawing, erasing, color selection, brush-size control, and canvas operations.
- Implemented freehand drawing using **pointer events** and **Polyline rendering** with configurable stroke properties.
- Built **Undo/Redo using stack-based state management** and designed the interface using AXAML, event handling, and data binding.

Design Devi – AI Design Consultant

Express.js · Groq API · JavaScript · Prompt Engineering

- Built an **AI-powered design consultant chatbot** providing guidance on UI/UX, branding, typography, accessibility, and design tools.
- Integrated the **Groq API** with an Express.js backend to process queries and generate real-time AI responses.
- Implemented server-side topic filtering and **persona-based prompt engineering** to improve response relevance.

Maximum Rectangle Fitting in a Pentagon

C++ · STL · CMake · Gnuplot

- Developed a C++ **geometry application** to determine and visualize rectangle fitting within an irregular pentagon.
- Designed reusable **Point**, **Shape**, **Rectangle**, and **Polygon** classes using OOP and SOLID principles.
- Implemented geometric calculations, STL containers, smart pointers, exception handling, and file I/O; integrated **CMake** and **Gnuplot** for build management and visualization.

E-commerce Backend API

Node.js · Express.js · MongoDB · JWT · REST APIs

- Built a **RESTful backend** supporting authentication, product management, cart operations, and order processing.
- Implemented **JWT authentication and role-based authorization** for customer and administrator access.
- Designed modular routes, controllers, middleware, and MongoDB models using **MVC architecture**, with search, filtering, sorting, and pagination APIs.